A02 NEURAL NETWORK ZOO

FOCUS: CNN CHEETAH 🐆

Introduction

What is a neural network?

Core idea

A neural network is a machine learning model made of artificial neurons connected in layers. Each neuron receives inputs, applies weights and a bias, and passes the result through an activation function before sending information forward.

Deep learning

Deep learning uses multiple hidden layers, and IBM notes that these hidden layers are where most of the learning occurs. Earlier layers learn broader patterns, while deeper layers learn more detailed features.

How neurons form a network

Input layer

Holds raw features such as pixels, words, or measurements before processing begins.

Hidden layers

Transform data into new representations through weighted connections, bias terms, and activation functions.

Output layer

Produces the final prediction, such as a class label, probability, or decision.

Turning networks into animals

The Neural Network Zoo is a visual way to organize neural network architectures, and it works well for class projects because different models have different behaviors and strengths.

🦁 One zoo, many architectures, each with a special role

Featured animals in the zoo

CNN Cheetah 🐆

Fast at spotting visual patterns such as edges, shapes, and objects in images.

RNN Raccoon 🦝

Moves through sequence data step by step while carrying context from earlier inputs.

LSTM Lemur 🐒

Uses memory cells and gates to keep important information for longer sequences.

Transformer Owl 🦉

Uses attention to focus on the most important parts of a sequence.

Autoencoder Beaver

Compresses information and rebuilds it to learn useful hidden features.

GAN Fox 🦊

Creates realistic new data through competition between generator and discriminator.

Why I chose CNN Cheetah

Fast visual hunter

A cheetah reacts quickly to visual targets, just as a CNN quickly detects patterns inside image data.

Sharp focus

CNNs focus on local parts of an image first, which fits the cheetah idea of targeting important details.

Strong match

This animal is easy to visualize and strongly connected to a real neural network used in practice.

How CNN Cheetah is built

Convolution layers

IBM describes the convolutional layer as the first layer of a convolutional network and the key feature detector in the model.

Pooling layers

Pooling layers reduce the size of the feature maps while keeping the most useful information for later stages.

Fully connected layers

IBM notes that the fully connected layer is the final layer, where the network uses learned features to make the final classification.

From simple features to full objects

🐆

IBM explains that earlier CNN layers focus on simple features such as colors and edges. As the image moves deeper through the network, later layers recognize larger elements, shapes, and finally the intended object.

Feature growth

Edges and lines
Early

Objects and classes
Middle

Textures and shapes
Late

Typical uses of CNN Cheetah

Real-world tasks

- Image classification
- Object recognition
- Medical image analysis
- Speech and audio pattern detection

Why CNNs are practical

IBM Developer describes CNNs as predominantly used for image recognition-based tasks. That makes CNN Cheetah a strong choice because it connects directly to one of the most recognizable and useful deep learning applications.

FINAL TIPS & TAKEAWAYS

- After exploring the neural network zoo, I can see that each model has its own strength. I chose **CNN Cheetah** because it fits image-based tasks really well, while RNN Raccoon and LSTM Lemur are better for sequence data, and Transformer Owl is stronger for language and attention-based tasks.
- This activity helped me understand that neural networks are not one size fits all. Each architecture is useful for a different kind of data, which is why choosing the right model matters.

THANK YOU

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